

Tide retrospective: *Sextic J-function asymptotic* *(grammar precursor, Step 3)*

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Abstract

For the sextic potential $K(w) = w^6/720$ and a continuous, globally bounded prior $\pi : \mathbb{R} \rightarrow \mathbb{R}$, the J-function $J(t) := \int_{\mathbb{R}} e^{-tK(w)} \pi(w) dw$ satisfies $t^{1/6} \cdot J(t) \rightarrow \pi(0) \cdot \frac{1}{3} \cdot 720^{1/6} \cdot \Gamma(1/6)$ as $t \rightarrow \infty$. This is the $k = 3$ mirror of the $k = 2$ quartic theorem landed earlier today (E4 / `tide/grammar-j-function`) and the third step of the grammar-paper precursor sequence. Two theorems in a new file `Laplace/OneD/JFunctionSextic.lean`: the centred DCT core `sextic_jfunction_centered_tendsto_zero` and the user-facing asymptotic `sextic_jfunction_asymptotic` as a two-line corollary via `sextic_partition`. 280 lines of Lean, 0 sorry, 0 axiom, 0 native_decide. Single-session run; `lake build` green on first compile; no GPT-5.5 Pro consult (the credentials file was missing in this session, and Candidate A was a direct mirror of E4 with the closed form numerically verified to machine precision).

1 Setting

The grammar paper (Murfet & Lehalleur, in preparation) needs a catalogue of single-component blow-up J-function asymptotics. Step 1 of the precursor sequence (Tide 7) closed the quartic bounded-prior partition. Step 2 (Tide 11+ / E4, today) closed the J-function asymptotic for the quartic potential $K(w) = w^4/24$ at the full-line scale, with a continuous-at-zero bounded prior. Step 3 — this Tide — mirrors Step 2 to $K(w) = w^6/720$, the next monomial in the sequence.

The seabed was `laplace` at `main` (commit 141997c), which already carried `Sextic.lean`’s closed-form partition, integrability, and moment lemmas:

$$\int_{\mathbb{R}} e^{-t \cdot x^6/720} dx = \frac{1}{3} \cdot \left(\frac{720}{t}\right)^{1/6} \cdot \Gamma\left(\frac{1}{6}\right),$$

plus `sextic_integrable_pow` (polynomial-times-Gibbs integrability for any $n \in \mathbb{N}$). E4’s `JFunctionQuartic.lean` was on the unmerged sibling branch `tide/grammar-j-function` but visible via `git show`; its proof template — substitute, then DCT — was the natural mirror target. The tide deliberation (`tide-pick`’s v2 survey of 2026-05-07) ranked Candidate L1 as the obvious top pick: cache-fresh after E4, single specific potential, single specific normalisation, and the only friction is constants ($4 \rightarrow 6$, $24 \rightarrow 720$, $1/4 \rightarrow 1/6$, $1/2 \rightarrow 1/3$).

The constant ratio is worth a sentence. The quartic case yielded $\frac{1}{2} \cdot 24^{1/4} \cdot \Gamma(1/4)$; the sextic yields $\frac{1}{3} \cdot 720^{1/6} \cdot \Gamma(1/6)$. The leading $\frac{1}{p}$ -vs- $\frac{1}{p}$ shift comes from Mathlib’s `integral_rpow_mul_exp_neg_mul_rpow`, which gives the half-line integral with a factor $1/p$ for $K = bx^p$; the full-line partition then doubles to $2/p$ ($\frac{1}{2}$ for $p = 4$, $\frac{1}{3}$ for $p = 6$). Numerical sanity at $\pi = 1$ matched the closed form to $\sim 10^{-15}$ at every t in the 0.1–1000 range.

2 The thing formalised

For a continuous prior $\pi : \mathbb{R} \rightarrow \mathbb{R}$ with global bound $|\pi(w)| \leq M$,

$$\lim_{t \rightarrow \infty} t^{1/6} \cdot \int_{\mathbb{R}} e^{-tw^6/720} \pi(w) dw = \pi(0) \cdot \frac{1}{3} \cdot 720^{1/6} \cdot \Gamma(\frac{1}{6}).$$

The Lean signature (`Laplace.OneD.sextic_jfunction_asymptotic`):

```
theorem sextic_jfunction_asymptotic
  {prior : ℝ → ℝ} (hprior_cont : Continuous prior)
  {M : ℝ} (hprior_bd : ∀ x, |prior x| ≤ M) :
  Tendsto (fun t : ℝ =>
    t ^ ((1:ℝ)/6) * ∫ w : ℝ, Real.exp (-(t * w^6 / 720)) * prior w)
    atTop (N (prior 0 * (1/3) * (720:ℝ) ^ ((1:ℝ)/6) * Real.Gamma (1/6)))
```

The load-bearing theorem is the centred form `sextic_jfunction_centered_tendsto_zero`, which states that $t^{1/6} \cdot \int e^{-tw^6/720} (\pi(w) - \pi(0)) dw \rightarrow 0$. This is the form the grammar paper will cite in its J-function asymptotic identity: the prior perturbation $\pi(w) - \pi(0)$ is asymptotically negligible against the concentrating Gibbs weight, and so the J-function is determined to leading order by $\pi(0)$ alone. The asymptotic-form corollary above is a two-line wrapper that splits $\pi(w) = (\pi(w) - \pi(0)) + \pi(0)$, applies the centred theorem to the perturbation, and applies `sextic_partition` (closed-form) to the constant piece, with the cancellation $t^{1/6} \cdot (720/t)^{1/6} = 720^{1/6}$ tying it together.

The hypotheses match E4: `Continuous prior` plus a global L^∞ bound. The mathematically minimal class is `AEMeasurable + ContinuousAt prior 0 + global bound`; we deliberately did *not* weaken in this tide, so the sextic and quartic files share an API surface. The hypothesis-weakening is E4 follow-up L3 and should be done as a single refactor across both files (and a `lean-common` substitution lemma) — landing it asymmetrically here would have created an awkward sextic-stronger quartic-weaker mismatch.

3 Proof strategy

The proof is substitute-then-DCT, identical in structure to E4 with the obvious constant rewrites.

The substitution. Set $a := t^{1/6}$, so $a^6 = t$. The change of variables $u = a \cdot w$ via Mathlib’s `Measure.integral_comp_mul_right` reshapes

$$t^{1/6} \int_{\mathbb{R}} e^{-tw^6/720} (\pi(w) - \pi(0)) dw$$

into

$$\int_{\mathbb{R}} e^{-u^6/720} (\pi(u/a) - \pi(0)) du.$$

The Jacobian bookkeeping ($a > 0$, $|a^{-1}| = 1/a$, $(w \cdot a)^6 = tw^6$, $(w \cdot a)/a = w$) is the load-bearing piece — about 50 of the file’s 280 lines. The cast $a^{(6:\mathbb{N})} = a^{(6:\mathbb{N}:\mathbb{R})}$ uses `Real.rpow_natCast`; the exponent identity $a^6 = (t^{1/6})^6 = t^{(1/6) \cdot 6} = t^1 = t$ uses `Real.rpow_mul` and `Real.rpow_one`.

The DCT. The substituted integrand $F(t, u) := e^{-u^6/720} \cdot (\pi(u/a) - \pi(0))$ has a t -independent envelope bound $(u) := 2M \cdot e^{-u^6/720}$, integrable from `sextic_integrable_pow 0` at $t = 1$. The point-wise limit is zero: as $t \rightarrow \infty$, $u/t^{1/6} \rightarrow 0$ (from `tendsto_rpow_atTop` at exponent $1/6 > 0$), and continuity of π at 0 gives $\pi(u/t^{1/6}) - \pi(0) \rightarrow 0$. Mathlib’s `tendsto_integral_filter_of_dominated_convergence` closes.

The asymptotic corollary. Split $\pi(w) = (\pi(w) - \pi(0)) + \pi(0)$ inside the integrand. The centred piece tends to zero by the headline theorem. The constant piece equals $\pi(0) \cdot \int e^{-tw^6/720} dw$, which by `sextic_partition` is $\pi(0) \cdot \frac{1}{3} \cdot (720/t)^{1/6} \cdot \Gamma(1/6)$. Multiplying by $t^{1/6}$, the $(720/t)^{1/6}$ factor combines with $t^{1/6}$ via $t^{1/6} \cdot (720/t)^{1/6} = 720^{1/6}$ (one `Real.mul_rpow` step plus `field.simp`).

The whole proof is therefore a two-step move: a 50-line substitution identity, then a four-clause invocation of DCT, then an integration- splitting corollary. No mathematical novelty over E4; the whole tide is a constants-substitution exercise, exactly as a Step-3-of-the- sequence tide should be.

4 Roadblocks and resolutions

There were no serious roadblocks. The file compiled green on first compile after the `lake exe cache get` step. The substitution identity, which absorbed most of E4’s effort, was a direct port — swap $\frac{1}{4}$ for $\frac{1}{6}$ throughout, swap a^4 for a^6 , swap `quarticPotential` for `sexticPotential`, and the `ring / field.simp` closures all fire identically.

The only friction was infrastructural: the worktree was created via the K1 worktree wrapper but did not inherit the canonical clone’s `.lake/build/` directory, so the first `lake build` attempt started compiling Mathlib from source. Killed and re-ran with `lake exe cache get` first, which restored the prebuilt oleans in ~ 15 seconds; the subsequent build took ~ 2 minutes (target file alone in 3.1 s). This is a one-line addition to the worktree wrapper: every fresh worktree should run `lake exe cache get` as part of `tide-worktree create`.

A second infrastructural friction: the GPT-5.5 Pro credentials file at `.agents/skills/timaeus-research/credentials` was missing from the session. Earlier today’s tides had access; somehow the file was removed or never copied locally. The deliberation proceeded on Claude judgment alone, justified by the fact that the candidate was a direct mirror of an already-deliberated proof and the closed-form constant was numerically verified at machine precision. This is acceptable for mirror-shaped tides; for any tide with substantive mathematical novelty, the credentials wall would need to be unblocked first.

5 What was Mathlib, what was new

Mathlib provided the entire infrastructure. Specifically: `MeasureTheory.Measure.integral.comp_mul_right` for the substitution; `Real.rpow_natCast`, `Real.rpow_mul`, and `Real.rpow_one` for the exponent arithmetic; `Real.mul_rpow` for the $t^{1/6} \cdot (720/t)^{1/6} = 720^{1/6}$ cancellation; `tendsto_rpow_atTop` for $t^{1/6} \rightarrow \infty$; `MeasureTheory.tendsto_integral_filter_of_dominated_convergence` for the DCT; `MeasureTheory.integral_add`, `MeasureTheory.integral_mul_const`, and `Integrable.const_mul / Integrable.mul_const` for the linearity bookkeeping.

What was new was, modulo constants, identical to what was new in E4: the substitution lemma `sextic_jfunction_centered_subst`, the bounded-dominator wrapper `sextic_dominator_integrable`, the DCT theorem, the asymptotic corollary. No genuinely new *mathematical* content appeared; the tide’s contribution is to extend the closed-form J-function template from $k = 2$ to $k = 3$ inside the same Lean infrastructure, which is the small step the grammar-precursor sequence needs to demonstrate the template generalises.

6 Lessons

- *Worktree wrapper should run lake exe cache get.* The K1 wrapper creates a worktree from `git worktree add` but does not seed `.lake/build/`. The fresh worktree has `.lake/packages/mathlib/.lake/build/`.

only partially populated (whatever was synced as part of clone), and a naive `lake build` re-compiles ~ 2700 Mathlib files. The fix is one line: append `lake exe cache get to tide-worktree create's` tail. (Promote to the `tide` skill's branch hygiene preflight too.)

- *Credentials hygiene.* The `.agents/skills/timaeus-research/credentials.json` file should be a session-startup invariant for any tide that may need GPT. Worth adding a one-line preflight check to the `tide` skill: if the credentials are missing, surface to the user before Step 2.
- *Mirror tides are cheap.* A direct constants-mirror of an already-deliberated tide can land in a single short session with no GPT consult, provided the closed form is numerically verified at the destination constants. Cost ratio for this tide vs E4 (both single-session, similar friction): roughly 1:3 in agent-time, attributable entirely to the absence of deliberation overhead and the inherited proof template.
- *The $1/p$ -vs- $2/p$ partition prefactor needs flagging.* The constant differs between $p = 4$ and $p = 6$ ($\frac{1}{2}$ vs. $\frac{1}{3}$), and a copy-paste agent that mirrors E4 mechanically without checking `sextic_partition's` actual right-hand side will land the wrong constant. The v2 survey did exactly this in its L1 entry (it copied $\frac{1}{2}$ from E4); the tide-log corrected the survey before formalising. Worth a flag in the `tide-pick` skill: when proposing a mirror candidate, surface the destination's closed-form partition explicitly so the survey doesn't carry a copy-paste constant.

7 Follow-ups

- *Generic- k abstraction (E4 follow-up L1, now ripe).* With both $k = 2$ and $k = 3$ in the file and identical proof structure, the abstraction over $K(w) = w^{2k}/(2k)!$ is the natural next step. Closed-form constant $\frac{1}{k} \cdot ((2k)!)^{1/(2k)} \cdot \Gamma(1/(2k))$, modulo factors. Estimated ~ 300 lines for the parametric form.
- *Asymptotic-equivalence wrapper $J(t) \sim \pi(0) \cdot Z(t)$ (E4 follow-up L2).* Now applies to two files (quartic, sextic). Probably worth a single shared ~ 15 -line corollary that takes the `Tendsto` ($\tau^{1/p}.J$) *formanddividesthroughbythepartitionfunction*.
- *Hypothesis weakening (E4 follow-up L3).* Defer until both quartic and sextic land on `main`; then refactor both files together to use `AEMeasurable + ContinuousAt 0 + global bound`, with a shared substitution lemma in `lean-common`.
- *Promote `jfunction_centered_subst` to `lean-common` (E4 follow-up L4).* Would shrink both quartic and sextic files by ~ 50 lines each. Best done together with the L3 hypothesis-weakening so both files are visited once.
- *Adopt the `worktree-cache-seeding` fix (this tide's lesson).* One-line addition to `tide-worktree create`; expect to land as a process tide.
- *`\leanref`-tag the grammar paper (E4 follow-up L5).* Once the grammar paper has TeX statements pinning the quartic and sextic J-function asymptotics, tag both with `\leanref` pinned to commit `be4608f` (this tide).